

THE UNIVERSITY OF ZAMBIA
Department of Mathematics and Statistics
MAT1110: Foundation Mathematics and Statistics for Social Sciences
Tutorial Sheet 3 (2020/2021)

1. Simplify each of the following:

(a) i. i^{11}
ii. i^{61}

iii. i^{102}
iv. i^{250}

v. i^{900}
vi. i^{64002}

(b) Determine which of the following functions are even, odd or neither odd nor even.

i. $f(x) = x^2 - 4x + 5$

v. $f(x) = 5 - 5x - 3x^2$

ii. $f(x) = 2x + 1$

vi. $f(x) = 3x^4 + x^2 - 7$

iii. $f(x) = x^2 + 4x - 10$

vii. $f(x) = \frac{x^2+4}{x^3-x}$

iv. $f(x) = \sqrt[3]{x}$

viii. $f(x) = \frac{3}{x^2+2}$

2. (a) Determine the nature of roots of each of the following equations:

i. $2x^2 - x + 5 = 0$

ii. $x^2 + 4x - 2 = 0$

iii. $x^2 - 4x + 4 = 0$

(b) For each of the following, find the values of k for which the equation has equal roots.

i.

$$x^2 + kx + 4 = 0$$

ii.

$$kx^2 + 8x + k = 0$$

(c) By completing the square, rewrite each of the following functions in the form

$$f(x) = a(x - h)^2 + k.$$

Hence, sketch the graph, showing clearly the x and y intercepts, and the turning point. State the minimum or maximum value of the function.

i. $f(x) = 2x^2 - 4x + 5$

v. $f(x) = 3x^2 + 6x - 4$

ii. $f(x) = 4 - 3x^2$

vi. $f(x) = 4x^2 - 12x + 3$

iii. $f(x) = 4x^2$

vii. $f(x) = x^2 + 10x + 25$

iv. $f(x) = -3x^2$

(d) Sketch on the same axes, the following pairs of functions, clearly showing the x and y intercepts, point(s) of intersection and the turning point.

i. $y = x^2 - 4x$, $x + y = 0$

ii. $y = 3x^2 - 2x$, $y = 1 - 4x$

iii. $y = x^2$, $y = -x^2 + 6x$

3. The number of *Vibrio cholerae* in refrigerated water is given by

$$N(T) = 20T^2 - 20T + 120, \text{ for } -2 \leq T \leq 14,$$

where T is the temperature of the water in degrees Celsius.

(a) Find the number of bacteria at 0° .

(b) Find the number of bacteria at 10° .

(c) At what temperature will the number of bacteria be minimal.

4. The profit function in dollars per day for a small scale company is given by

$$f(x) = -16x^2 + 64x + 190,$$

where x is the number of products sold each day. Find the number of products (x) sold each day in order to maximize the profit, and find this maximum profit.

5. The length of a rectangular fence is three more than twice the width. Determine the dimensions that will give the total area of $27m^2$.

6. (a) Use **long division** to divide each of the following polynomials by the given divisor.

i. $f(x) = x^2 + 3x + 5,$ $x + 1$

ii. $f(x) = 6x^3 - 19x^2 + 16x - 4,$ $x - 2$

iii. $f(x) = x^3 + 6x^2 - x + 3,$ $2x^2 - 1$

iv. $f(x) = x^4 + 3x^2 + 1,$ $x^2 - 2x + 3$

(b) Use **synthetic division** to divide each of the following polynomials by the given divisor, and write the polynomial in the form

$$p(x) = q(x)(x - a) + r,$$

where $q(x)$ is the quotient and r is the remainder.

i. $p(x) = x^3 - 7x^2 - 4x + 28,$ $x + 2$

ii. $p(x) = 5x^3 - 6x^2 - 28x - 2,$ $x + 2$

iii. $p(x) = x^3 - 10x^2 - 31x - 30,$ $x - 3$

iv. $p(x) = 6x^3 + x^2 - 21x - 10,$ $2x + 1$

(c) Factorize completely each of the following polynomials:

i. $4x^3 - 8x^2 - x + 2$

iii. $x^3 - 2x^2 + 2x - 4$

v. $x^3 + 1$

ii. $2x^3 + 3x^2 - 3x - 2$

iv. $x^3 - 1$

vi. $x^4 - 6x^3 - 11x^2 + 24x - 28$

7. Solve each of the following equations:

(a) $x^3 + x^2 - x - 1 = 0$

(d) $x^2(2x + 1) = 13x - 6$

(b) $2x^3 + 3x^2 - 3x = 2$

(c) $x^3 + 6x^2 + 5x - 2 = 0$

(e) $x^4 - 6x^3 = 11x^2 - 24x - 28$

8. (a) Given that $x - 1$ and $x + 1$ are factors of $f(x) = ax^3 + bx^2 - 3x - 7$, find the value of a and of b .

(b) Let $f(x) = x^3 + ax^2 + bx + 6$. Given that the remainders when $f(x)$ is divided by $x + 1$ and $x - 2$ are 20 and 8 respectively, find the value of a and of b .

9. Sketch each of the following functions, clearly indicating the x and y intercepts.

(a) $f(x) = x^3 - 13x + 12$

(d) $f(x) = 4x^3 - 12x^2 + 5x + 6$

(b) $f(x) = 15 + 5x - 3x^2 - x^3$

(c) $f(x) = x^3 - 4x^2 + x + 6$

(e) $f(x) = x^3 - 6x - 5$